Project Proposal: Smart Recruitment & Resume Screening System (SRRSS)

1. Proposed Project Title:
Smart Recruitment & Resume Screening System (SRRSS)-An AI-Driven Enterprise Talent Acquisition Platform

2. Team Members:

Name	Student ID	Role
Love Shah	202512058	**Backend Architect:** API Logic, Workflow Engine & Database Design
Saiyad Mohammad Fahim	202512103	**Frontend Lead:** React UI, Dashboards & User Experience
Jagrat Jani	202512119	**AI/ML Engineer:** Resume Parsing, Scoring Algorithms & Python Microservice
Manav Vyas	202512062	**DevOps & QA:** Authentication, Access Control (RBAC), Documentation & Testing

3. List of Functionalities The system is designed as a mid-size enterprise application implementing a strict 3-tier architecture.

A. Presentation Tier (Client-Side)

- **Candidate Portal:** Secure login, drag-and-drop resume upload, and real-time application status tracking.
- **Recruiter Dashboard:** Job posting management, ranked candidate views, and interview scheduling interface.
- **Admin Console:** User management and system configuration.

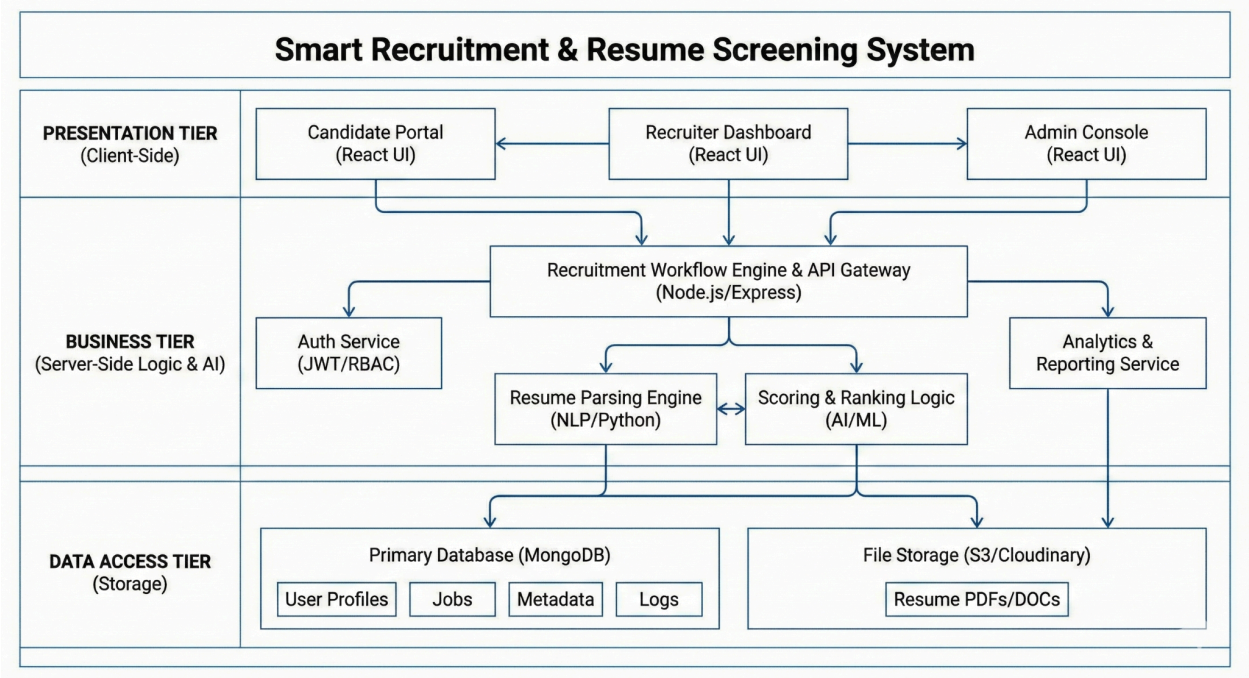
B. Business Logic Tier (Server-Side)

- **Automated Resume Parsing:** Extracts structured data (Skills, Education, Experience) from PDF/DOCX files.
- **Contextual Scoring Engine:** Ranks candidates against job descriptions using rule-based logic (v1) and NLP similarity (v2).
- **Recruitment Workflow Manager:** Manages state transitions (Applied → Shortlisted → Interview → Hired).
- **Bias Reduction:** Hides PII (Personally Identifiable Information) during initial screening to ensure fair evaluation.

C. Data Access Tier (Storage)

- **Structured Storage:** Stores user profiles, job metadata, and application logs.
- **File Storage:** Secure management of resume documents.
- **Audit Logging:** Tracks all recruiter actions for compliance and reporting.

4. Block Diagram Visualizing Workflow



5. Tentative Work Distribution

- **Member 1 (Backend):** Setup of Node.js/Express server, design of RESTful APIs for job management, and implementation of the MongoDB schema design.
- **Member 2 (Frontend):** Development of React.js components, integration of Tailwind CSS for responsive design, and state management (Redux/Context API).
- **Member 3 (AI/Logic):** Implementation of resume text extraction (using libraries like pdf-parse or Python spaCy), development of the scoring algorithm, and data analytics logic.
- **Member 4 (Security & Docs):** Implementation of JWT Authentication, Role-Based Access Control (RBAC) middleware, Unit Testing (Jest), and maintenance of Project Documentation.

6. Technologies to be Used

- **Presentation Tier:** React.js, HTML5, CSS3 (Tailwind), Axios.
- **Business Tier:** Node.js, Express.js (Main Logic), Python (Data Processing Microservice).
- **Data Access Tier:** MongoDB (NoSQL Database).
- **Tools:** Git/GitHub (Version Control), Postman (API Testing), Trello/Jira (Agile Tracking).